BPM 1

FTM: Lecture 38

Physiology of Autonomic Transmission

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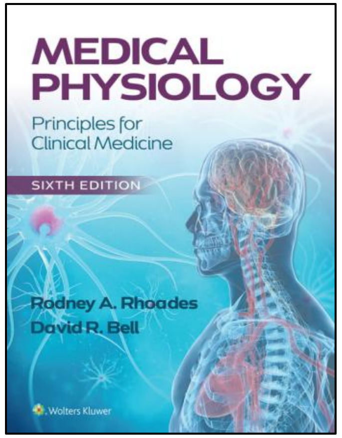
Recommended Reading

Chapter 6: Autonomic nervous System. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.



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Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

LEARNING OBJECTIVES

SOM.MK.II.BPM1.1.FTM.3.PHYS.0040 Describe the main functions of the Sympathetic Nervous System (SNS) and the Parasympathetic Nervous System (PNS)

SOM.MK.II.BPM1.1.FTM.3.PHYS.0041 Classify autonomic nerves based on the primary neurotransmitters released from their terminals.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0043 Describe the processes involved in cholinergic and adrenergic transmission.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0044 Describe the location and function of cholinergic and adrenergic receptor subtypes and the signaling mechanisms associated with them.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0045 Explain the concepts of autonomic tone, dual innervation, and predominant tone in the target tissues of the autonomic nervous system

Lecture Outline

- ❖Main functions of the autonomic nervous system (ANS)
- ❖Classification of autonomic nerves
- ❖Cholinergic and adrenergic transmission
- ❖Cholinergic and adrenergic receptors, and signaling mechanisms
- ❖Dual innervation and predominant tone of ANS

- NERVOUS SYSTEM
  - Central Nervous System
    - Brain
    - Spinal Cord
  - Peripheral Nervous System
    - Sensory Neurons
    - Motor Neurons
      - Somatic Nervous System
      - Autonomic Nervous System (ANS)
        - Sympathetic Nervous System
        - Parasympathetic Nervous System

Useful Definitions

Adrenergic, noradrenergic	A nerve ending that releases norepinephrine as the primary transmitter; also, a synapse in which norepinephrine is the primary transmitter
Adrenoceptor, adrenergic receptor	A receptor that binds, and is activated by, one of the catecholamine transmitters or hormones (norepinephrine, epinephrine, dopamine) and related drugs
Baroreceptor reflex	The homeostatic mechanism that maintains a constant mean arterial blood pressure
Cholinergic	A nerve ending that releases acetylcholine; also, a synapse in which the primary transmitter is acetylcholine
Cholinoceptor, cholinergic receptor	A receptor that binds, and is activated by, acetylcholine and related drugs
Dopaminergic	A nerve ending that releases dopamine as the primary transmitter; also, a synapse in which dopamine is the primary transmitter
Nonadrenergic, noncholinergic (NANC) system	Nerve fibers associated with autonomic nerves that release any transmitter other than norepinephrine or acetylcholine
Parasympathetic	The part of the autonomic nervous system that originates in the cranial nerves and the sacral part of the spinal cord
Sympathetic	The part of the autonomic nervous system that originates in the thoracic and lumbar parts of the spinal cord
Postsynaptic receptor	A receptor located on the distal side of a synapse, for example, on a postganglionic neuron or an autonomic effector cell
Presynaptic receptor	A receptor located on the nerve ending from which the transmitter is released into the synapse; modulates the release of transmitter

Lecture Outline

- **Main functions of the autonomic nervous system (ANS)**
- Classification of autonomic nerves
- Cholinergic and adrenergic transmission
- Cholinergic and adrenergic receptors
- Dual innervation and predominant tone of ANS

Autonomic Nervous System: Primary Effectors & Motor functions

- **Smooth muscle:**
  - BLOOD VESSELS → Blood pressure; Body temperature
  - GASTROINTESTINAL TRACT → Digestion
  - BLADDER → Micturition
  - BRONCHIAL TREE → Bronchial constriction/dilation
  - EYE (upper eyelid, iris, ciliary body) → Pupillary dilation/constriction
- **Heart** → Heart rate; Contractility; Conduction velocity
- **Glands**
  - **Endocrine glands** → Metabolism
    - PANCREAS
    - ADRENAL GLANDS (CORTEX)
  - **Exocrine glands**
    - LACRIMAL GLANDS → Lacrimation
    - SWEAT GLANDS → Sweating
    - SALIVARY GLANDS → Salivation

ANS controls involuntary functions of the body

Autonomic Nervous System: Motor functions

SYMPATHETIC

- **"Fight or Flight"**

To combat with Stress

- Increase in blood flow to the skeletal muscles
- Increase in heart rate
- Increase in blood pressure
- Increase in blood glucose level
- Pupillary dilation (mydriasis)

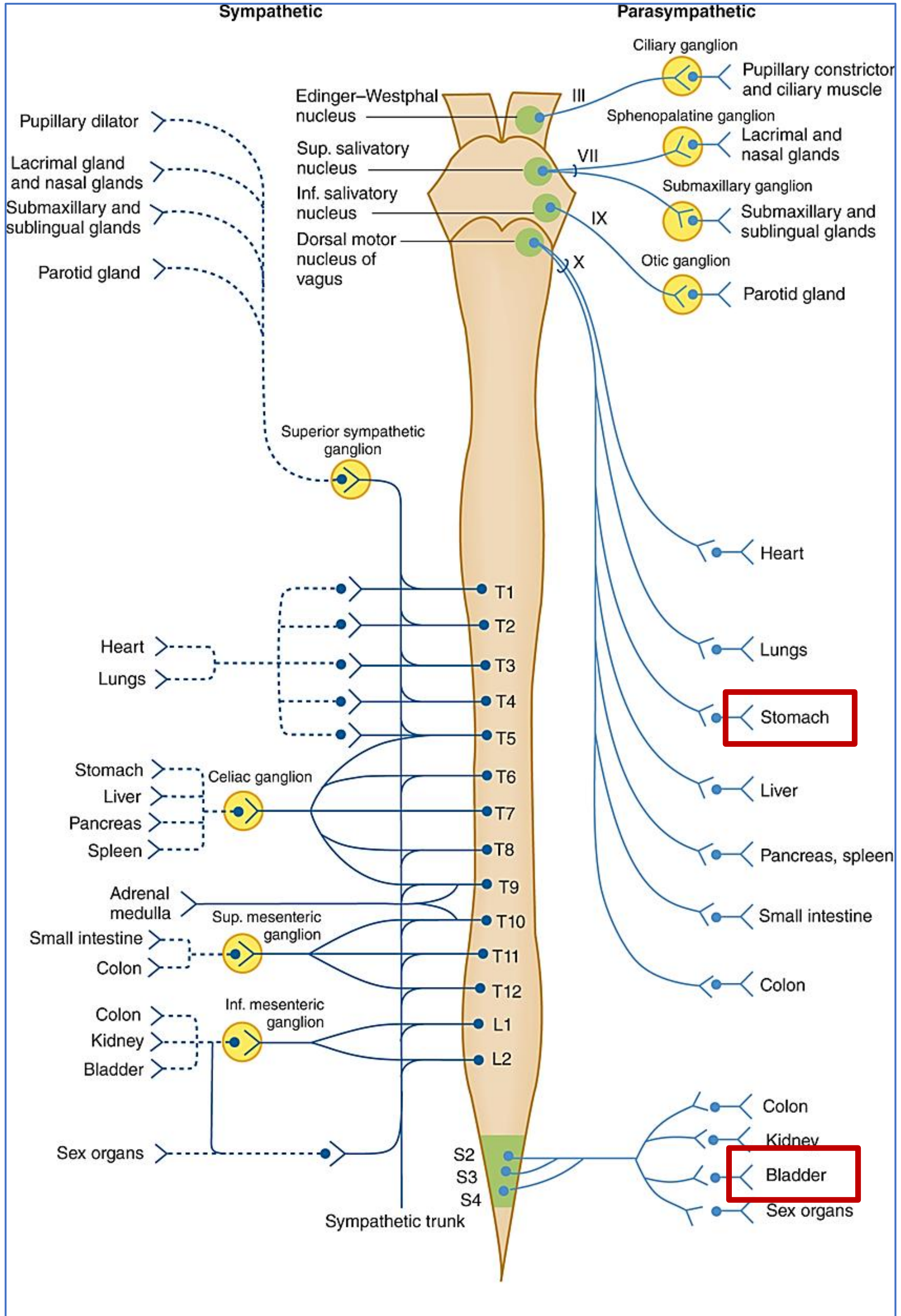
PARASYMPATHETIC

- **"Rest and Digest"**

Conservation and restoration of body energy

- Decrease in heart rate
- Increase in activity of gastrointestinal tract
- Pupillary constriction (miosis)

TWO DIVISIONS OF ANS



Injury/damage to one or more nerves affect organ/system functions.
Example:
Patient with diabetic neuropathy may have difficulty in controlling bladder or maintaining blood pressure.

Over-stimulation of one or more nerves may affect organ/system functions.
Example:
Chronic stimulation of vagus nerve may increase acid production and ulcers in the stomach.

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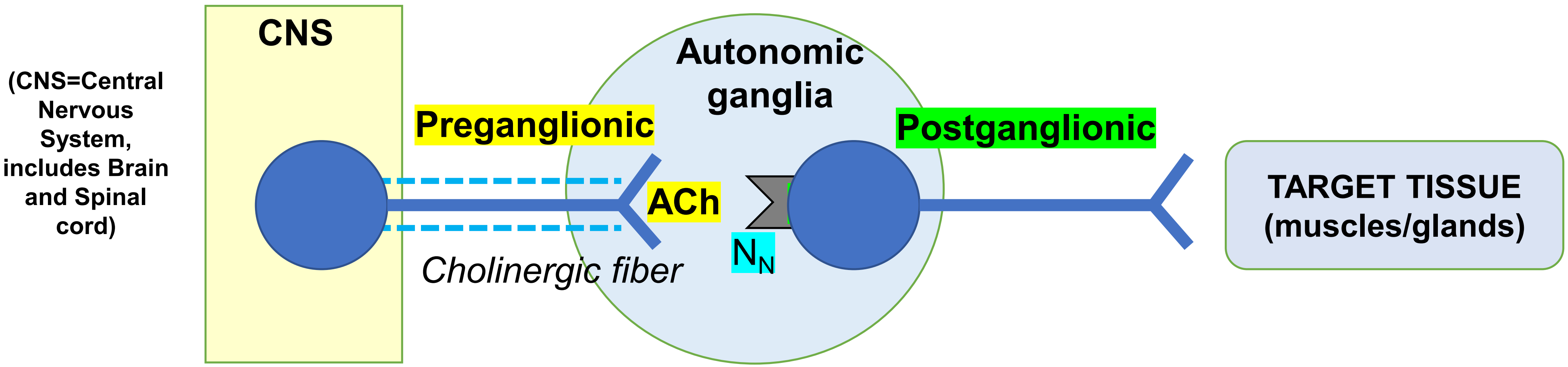
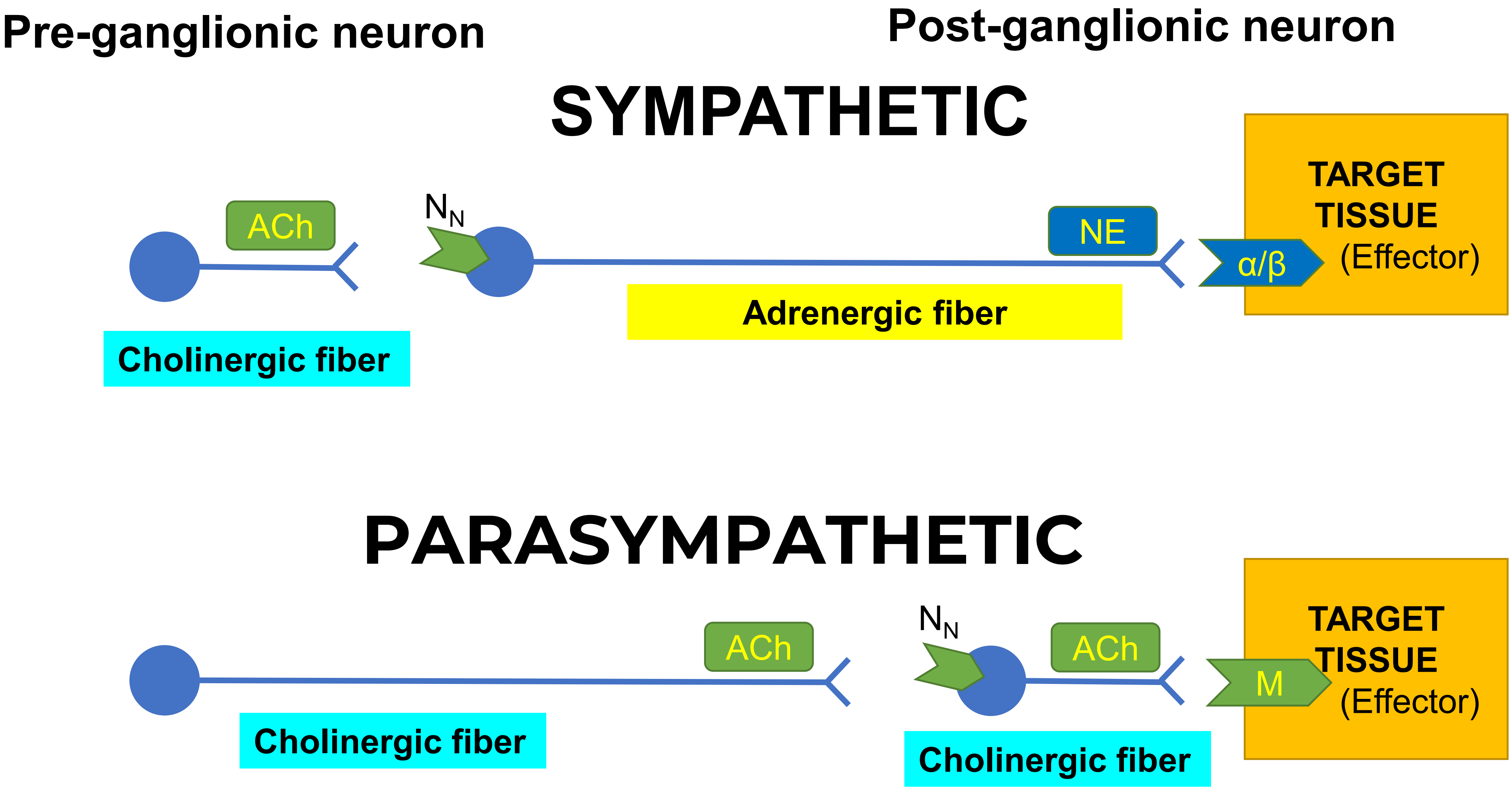
Common Characteristics Of Autonomic Nervous system



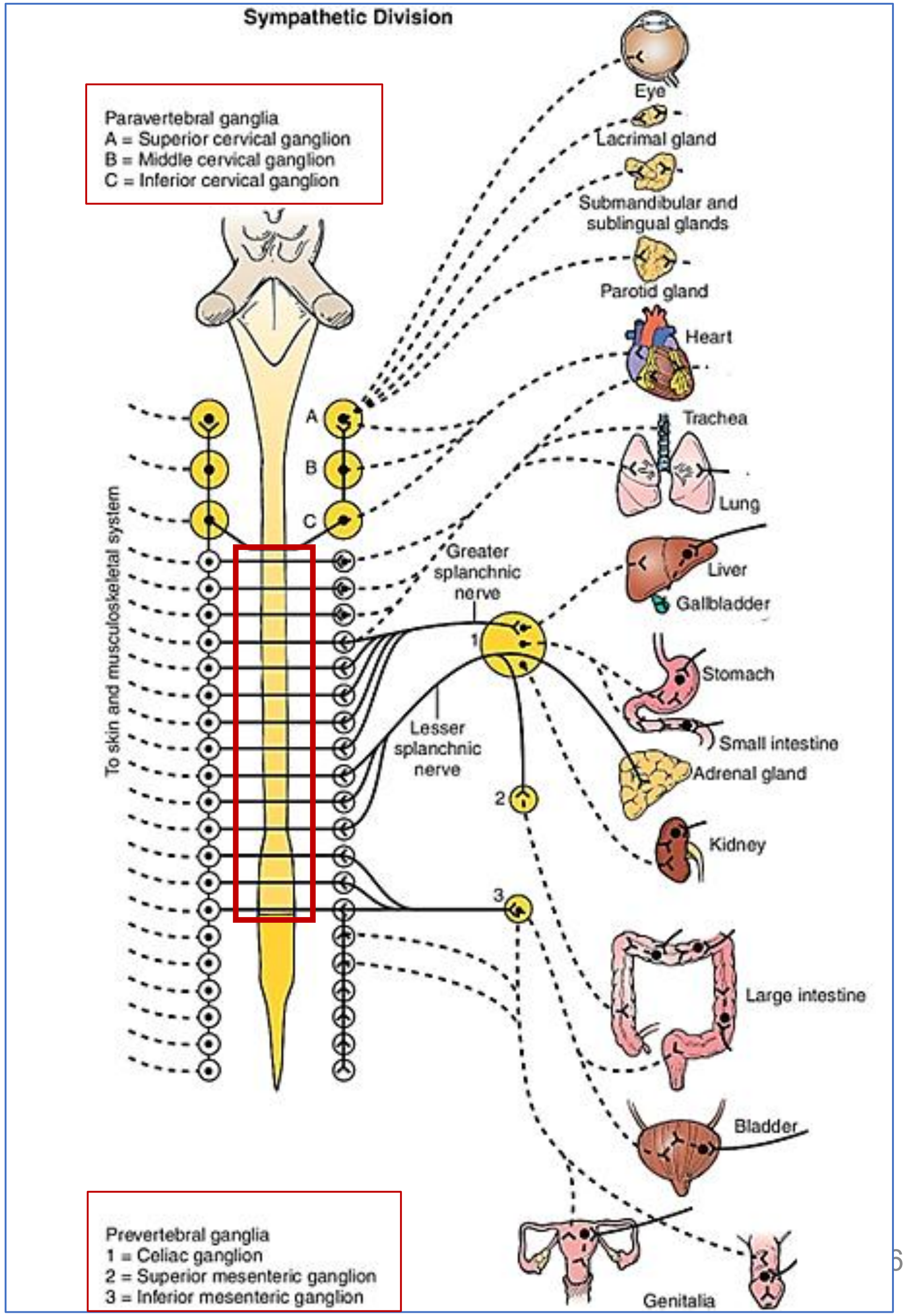
- Disynaptic pathway
- Preganglionic neuron is myelinated (faster transmission)
- Origin of preganglionic neuron is in the central nervous system (brain/spinal cord)
- Origin of the postganglionic neuron is in the autonomic ganglia, in the peripheral NS
- Preganglionic neuron releases **ACETYLCHOLINE (Ach)** as neurotransmitter, which binds to cholinergic nicotinic receptors (nicotinic subtype, N_N)



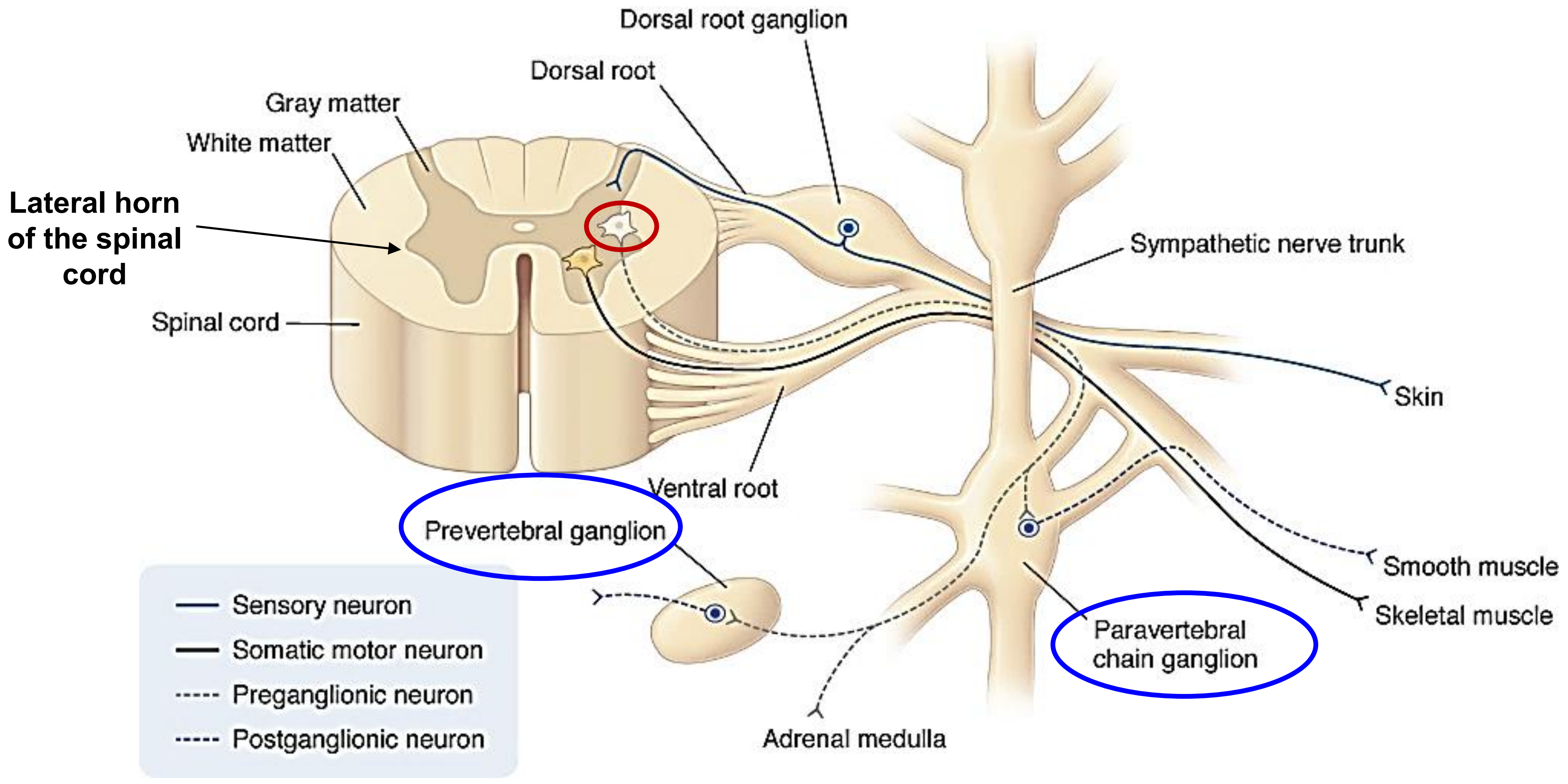
ACh: acetylcholine, NE: norepinephrine, M: muscarinic receptor, α/β: adrenergic receptors

SYMPATHETIC Nervous system: Organization

- ❑ Origin: **T1-L2** middle region of the spinal cord (**Thoracolumbar division**)
- ❑ Short preganglionic neurons
- ❑ Long postganglionic neurons
- ❑ Sympathetic ganglia located near CNS
- ❑ Ratio of preganglionic to postganglionic fibers is about 1:20



Organization of the peripheral nervous system



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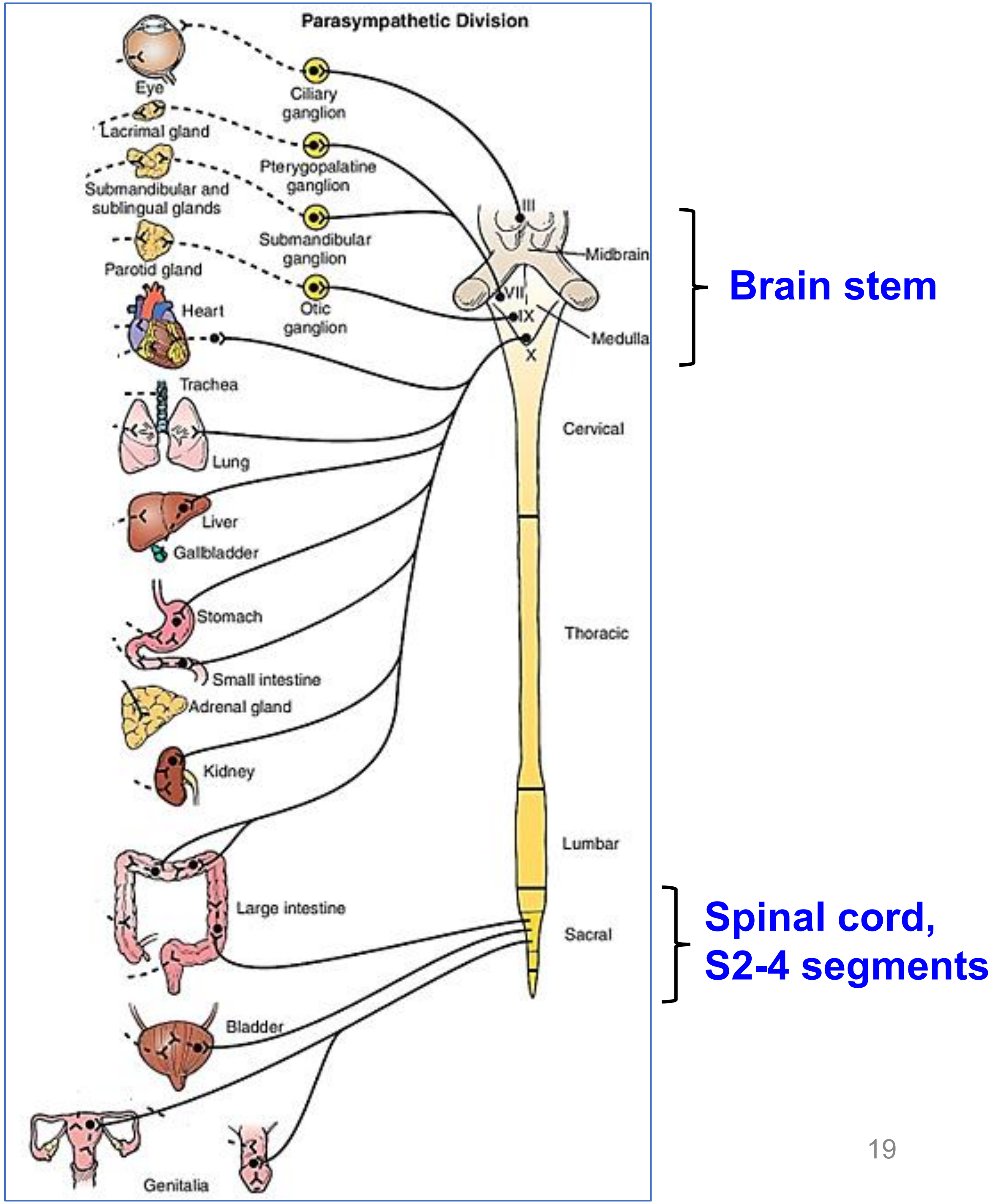
Organization of the peripheral nervous system: Explanation for previous image

The peripheral nervous system contains sensory, somatic motor, and autonomic components. Sensory neurons arise in the skin or joints, have cell bodies and nuclei in the **dorsal root ganglia**, and project onto neurons located in the dorsal horn of the spinal cord. Somatic motor neurons arise in the ventral horn of the spinal cord, exit through the ventral roots, and join fibers of sensory neurons to form **spinal nerves**, which then innervate skeletal muscle. The **autonomic component** of the peripheral nervous system consists of a two-nerve system: the two nerves are called preganglionic and postganglionic neurons, respectively. Sympathetic preganglionic neurons arise in the intermediolateral horn of the thoracic and lumbar segments of the spinal cord and project onto postganglionic neurons in the **paravertebral and prevertebral ganglia**. Sympathetic postganglionic neurons innervate many organs, including smooth muscle. The adrenal medulla is also innervated by preganglionic neurons of the sympathetic nervous system.

PARASYMPATHETIC Nervous system: Organization

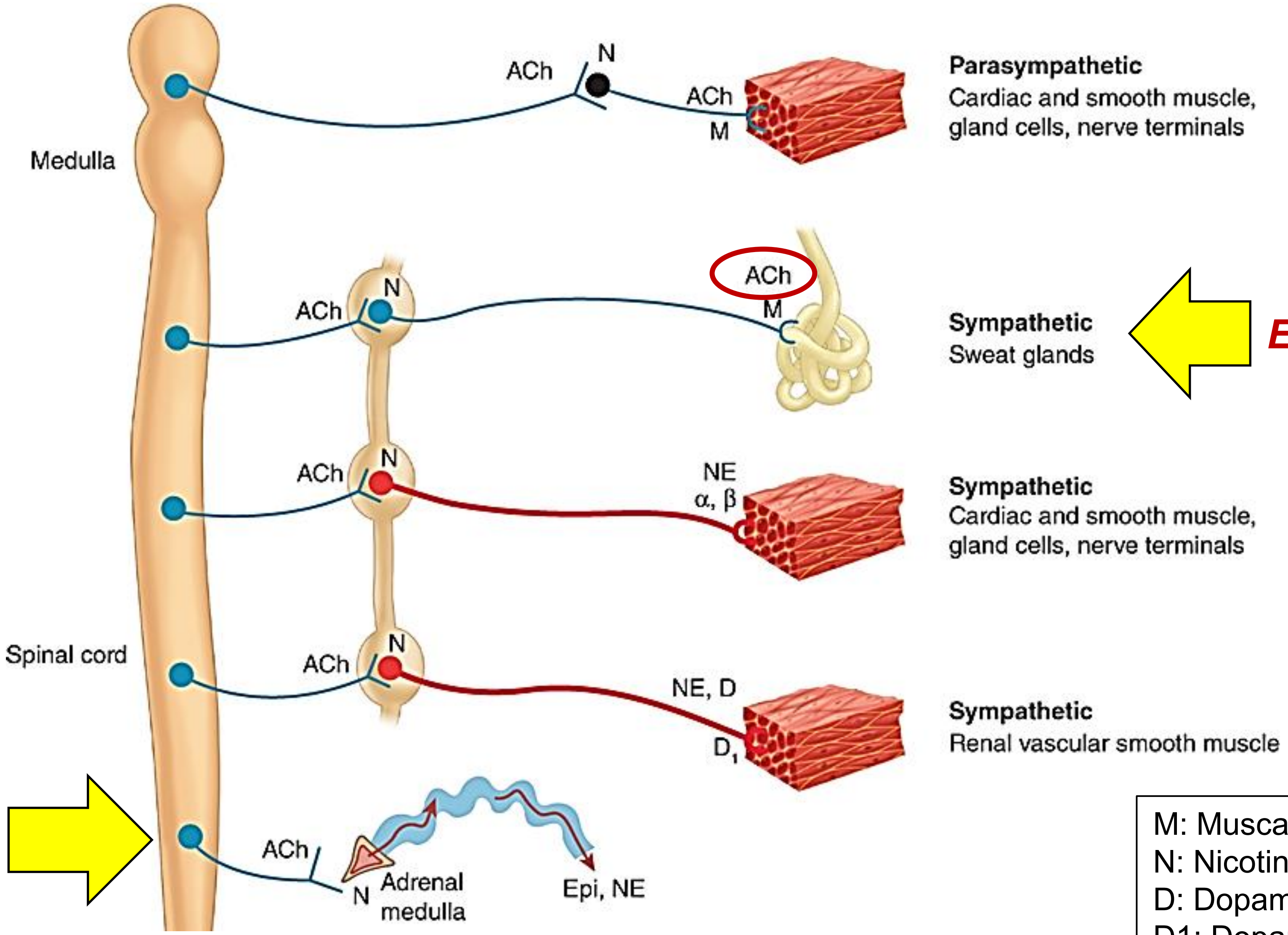
- ❑ Origin: **Brainstem & S2-S4** region of the spinal cord
- ❑ Long preganglionic neurons
- ❑ Short postganglionic neurons
- ❑ Parasympathetic ganglia located far from CNS
- ❑ Ratio of preganglionic to postganglionic fibers is about 1:3



Distinguishing Features of the Sympathetic and Parasympathetic Divisions

SYMPATHETIC	PARASYMPATHETIC
Origin: **T1-L2** spinal cord segments (lateral horn)	Origin: **brain stem** (autonomic motor nuclei of III, VII, IX, and X cranial nerves), and sacral spinal cord segments **S2-S4**
Autonomic ganglia location: • **Paravertebral sympathetic ganglion chain** • **Collateral ganglia** (prevertebral)	Ganglia located near or embedded within the target tissue
Short cholinergic preganglionic fiber	**Long** cholinergic preganglionic fiber
Long adrenergic postganglionic fiber	**Short** cholinergic postganglionic fiber
Ratio of preganglionic fibers to postganglionic fibers is **1:20**	Ratio of preganglionic fibers to postganglionic fibers is **1:3**
Activity often involves **massive discharge** of the entire system	Activity normally to **discrete organs**
Primary neurotransmitter of the postganglionic neurons is **norepinephrine**	Primary neurotransmitter of the postganglionic neurons is **acetylcholine**

“Exceptions” to the general rule in ANS



General rule

Exception!

General rule

Exception!

Exception!

M: Muscarinic
N: Nicotinic
D: Dopamine
D1: Dopaminergic-1
Epi: Epinephrine

Exceptions to “the rule”

ADRENAL MEDULLA:

Directly innervated by **preganglionic sympathetic fibers**. Post-ganglionic neurons secrete E and NE directly to the blood stream.

SWEAT GLANDS:

Innervated by **cholinergic sympathetic postganglionic neurons**

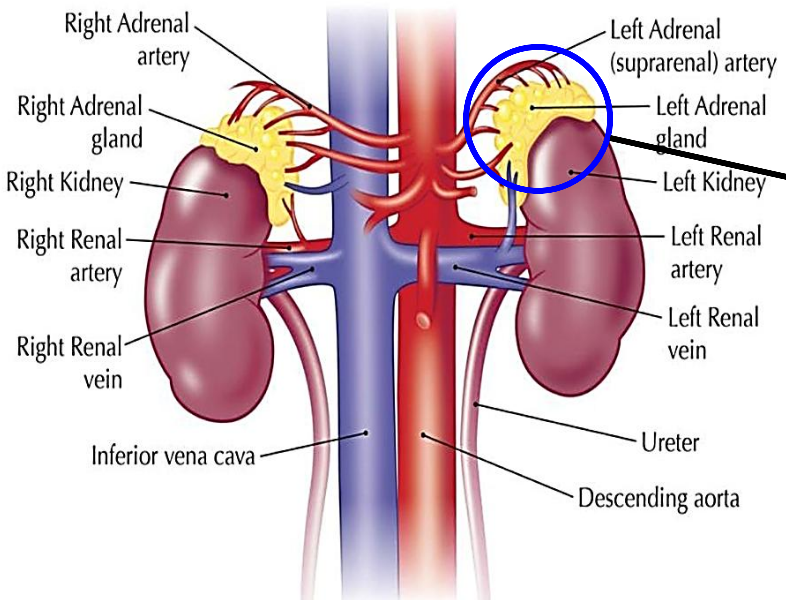
Receptors on the sweat glands are cholinergic muscarinic: **M3**

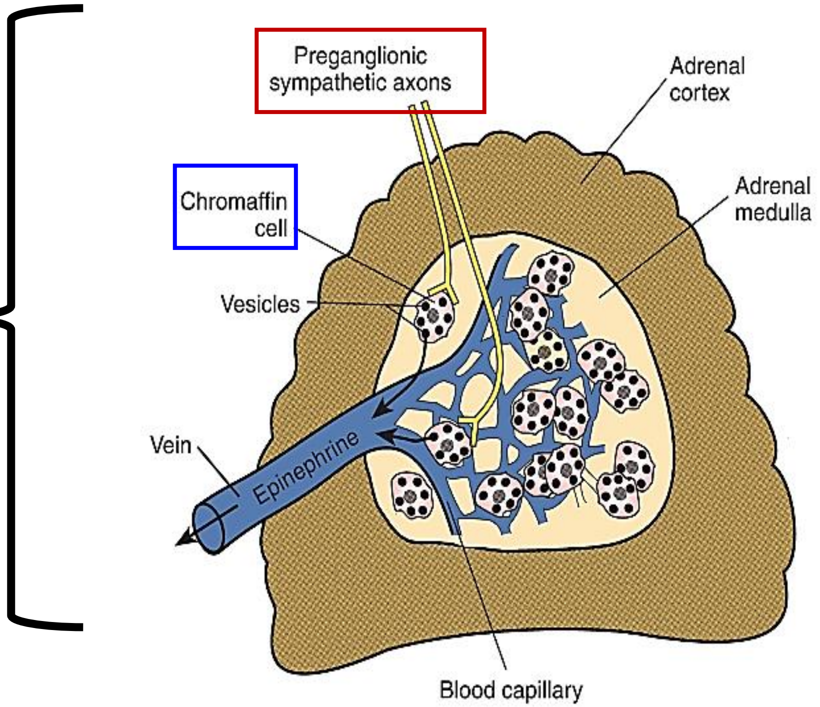
KIDNEY: Post-ganglionic sympathetic nerves secrete both NE and Dopamine as neurotransmitters.

Dopaminergic sympathetic postganglionic neurons supply to the renal vascular smooth muscles.

Receptors on the renal vascular smooth muscles are dopaminergic: **D1**

Adrenal Medulla functions as a Sympathetic ganglion





Preganglionic sympathetic axons synapse on chromaffin cells: **Ach binds to N_N** receptors.

Stimulation of the sympathetic nerves release Epinephrine (E) & Norepinephrine (NE) **into CIRCULATION:** 80% Epinephrine (E), 20% Norepinephrine (NE).

***Pheochromocytoma** is a malignant tumor of the sympathetic ganglia*

PHEOCHROMOCYTOMA

Fig: MRI scan showing a large left suprarenal mass (arrows)

E, NE and DA together called **Catecholamines**

- A tumor of the adrenal medulla
- Abnormal growth of **chromaffin cells**
- Tumor cells secrete epinephrine (E), norepinephrine (NE), and dopamine (DA) into circulation in spells
- **Symptoms**: Patient presents with sudden episodes of :
  - Elevated blood pressure
  - Headache
  - Excessive sweating (diaphoresis)
  - Palpitations and tachycardia
  - Pallor
- **Lab**: Elevated catecholamines and their metabolites in blood and urine
- May be associated with **familial syndromes** like Multiple endocrine neoplasia (MEN) syndrome and von Hippel-Lindau (VHL) disease.

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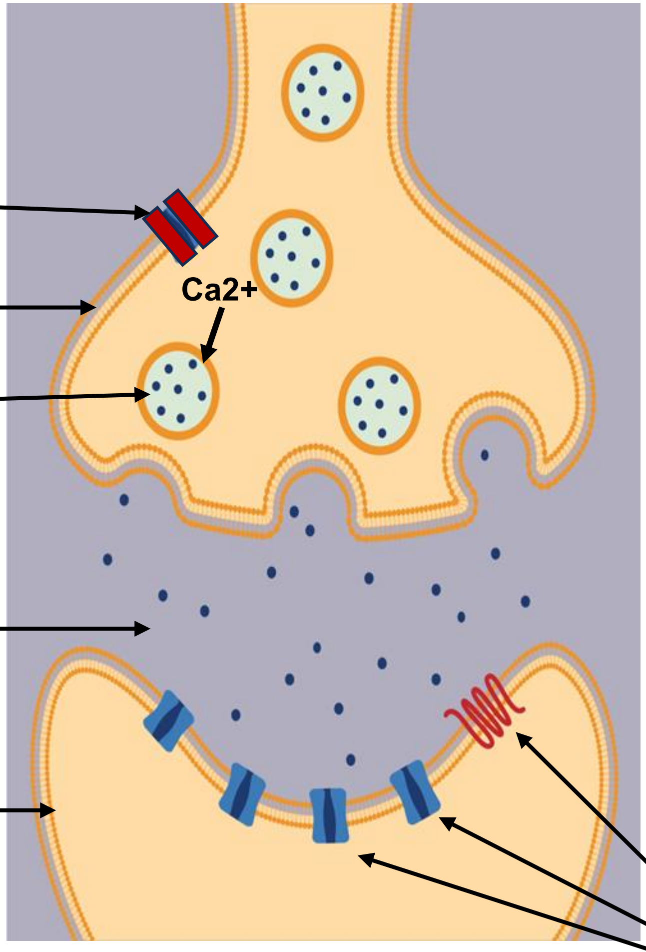
Cholinergic and Adrenergic Transmissions



Arrival of an action Potential at presynaptic terminal

↓

Opens Calcium channels

↓

Influx of calcium

↓

Calcium binds with synaptic vesicles

↓

Release of neurotransmitters

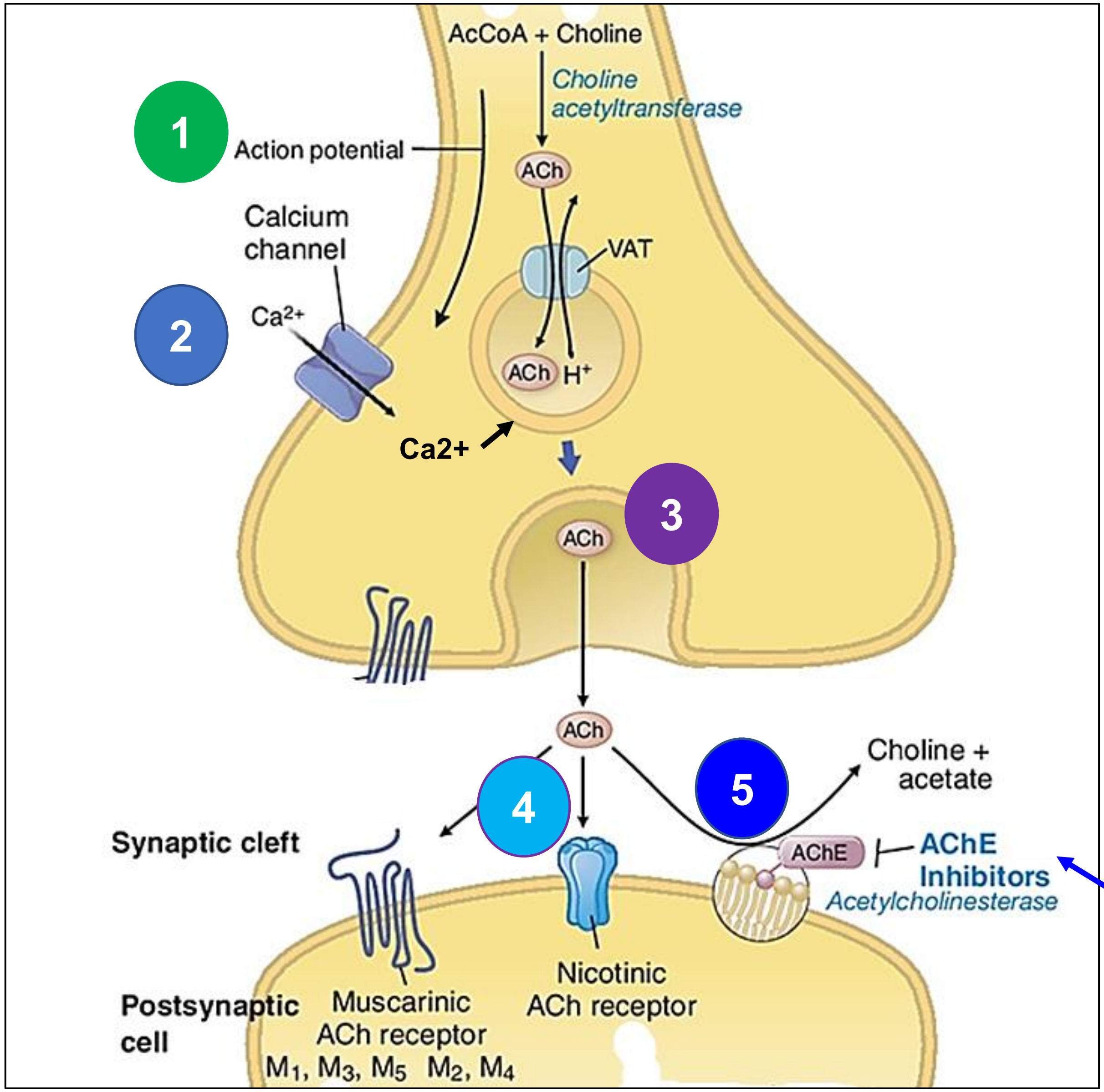
↓

Neurotransmitters bind with postsynaptic receptors

↓

Response in postsynaptic cell

CHOLINERGIC TRANSMISSION: Pre and Postsynaptic events



VAT: Vesicular ACh Transporter

1. Action potential at presynaptic terminal
2. Calcium influx
3. ACh release into synaptic cleft
4. ACh binds to postsynaptic cholinergic receptors
5. ACh action is terminated by enzyme, Acetylcholinesterase (AChE)

Insecticides work as AChE inhibitors!

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CHOLINERGIC TRANSMISSION: Pre and postsynaptic events

Events at Presynaptic Nerve terminal

1. At rest*: ACh is synthesized in the cytoplasm from **acetyl-CoA (AcCoA) and Choline**, a reaction catalysed by **choline acetyltransferase.** Ach is transported by the vesicular ACh transporter (VAT) and stored in the presynaptic vesicles.
2. During presynaptic nerve stimulation: ACh is released when an **action potential** depolarizes the terminal. Action potential triggers calcium influx through **voltage-gated calcium channels.** The calcium mediates Ach release.

Events in Synaptic Cleft and at Postsynaptic cell

Released Ach has 2 important destinations:

1. ACh binds to **cholinergic receptors (**either **nicotinic** or **muscarinic)** in the postsynaptic cell. This binding will either stimulate or inhibit postsynaptic cell.
2. An enzyme called **Acetylcholinesterase (AChE)** in the synaptic cleft splits ACh into choline and acetate (terminates ACh action).

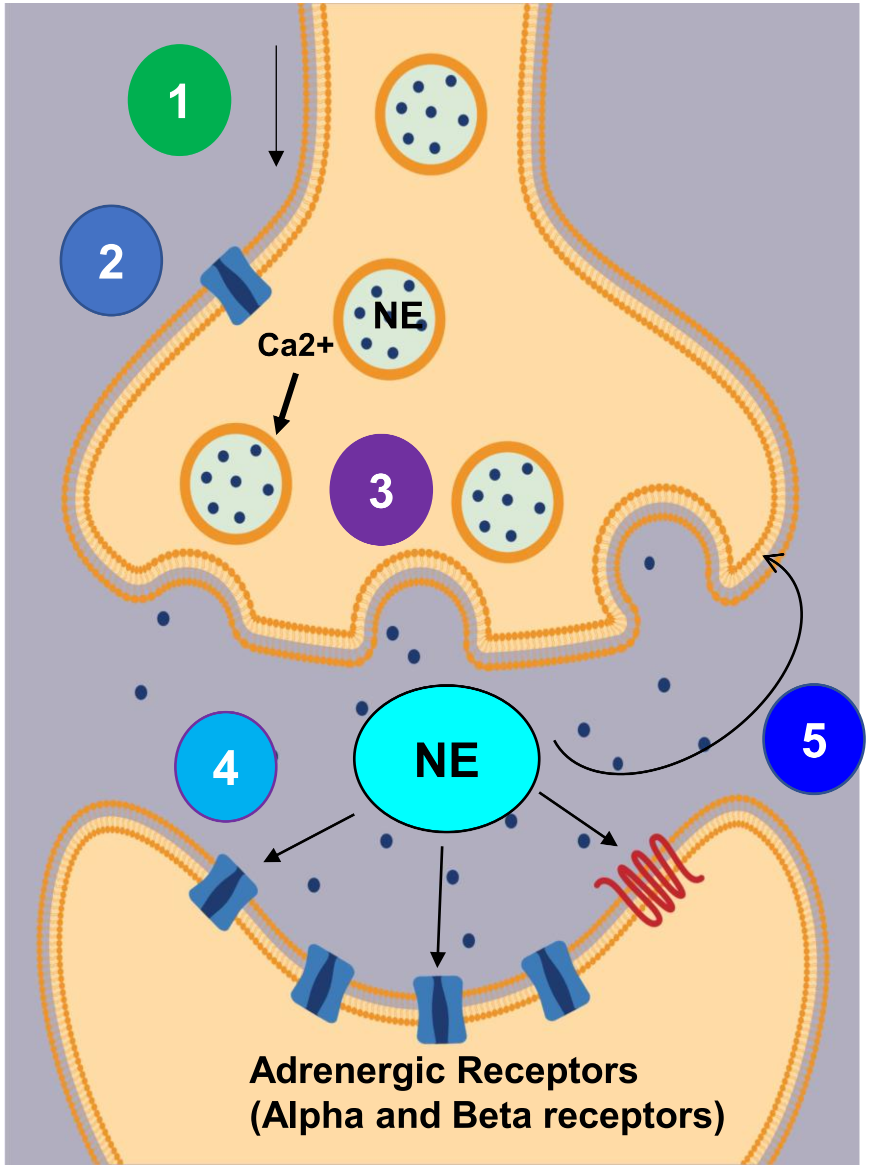
*(*At rest means when the presynaptic nerve is NOT stimulated)*

Organophosphate compounds: AChE (*acetylcholinesterase) enzyme inhibitors*

- Examples: Organophosphates used as insecticides (**Chlorothion, Malathion, Parathion)** and nerve gas agents.
- Organophosphates inhibit **irreversibly** the enzyme **AchE** and thus **increase Ach level** in the organs and tissues.
- **Signs and symptoms**: Patients with poisoning develop clinical features of elevated Ach everywhere in the body such as:
    - Muscarinic Receptor stimulation effects: diarrhea, urination, miosis, bronchospasm, bradycardia, emesis, lacrimation, sweating, salivation.
    - Nicotinic R stimulation effect: neuromuscular blockade and skeletal muscle paralysis.
    - CNS effects: respiratory depression, lethargy, seizures, and coma.

Signs and symptoms are like stimulation of the parasympathetic nervous system to the whole body

ADRENERGIC TRANSMISSION: Pre and Postsynaptic events



1. Action potential at presynaptic terminal
2. Calcium influx
3. Norepinephrine (NE) release into synaptic cleft
4. NE binds to postsynaptic adrenergic receptors (alpha1, beta1 and 2)
5. NE is also taken up by the presynaptic terminal

ADRENERGIC TRANSMISSION: Pre and Post synaptic events

Events at Presynaptic Nerve Terminal

1. At rest: The neurotransmitter, **Norepinephrine (NE)** is synthesized and stored in the synaptic vesicles.

2. During presynaptic nerve stimulation: NE is released when an action potential depolarizes the terminal. Action potential triggers calcium influx through voltage-gated calcium channels and the calcium mediates NE release.

Events in Synaptic Cleft and Postsynaptic cell

Released NE has 2 important destinations:

1. NE binds **adrenergic receptors, alpha or beta** in the postsynaptic cell.

2. NE also diffuses away from the postsynaptic receptor sites and is taken back into the presynaptic nerve terminal (terminates NE action).

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Major receptor systems in the ANS include:

- ❖ **Cholinergic receptors**
- ❖ **Adrenergic receptors**
- ❖ **Dopamine receptors**

Usually combined together as they are related to sympathetic nervous system

Cholinergic Receptors

Nicotinic Receptors
(Ion channel Linked)

- Muscle type N_M
- Neuronal type N_N

Ligand-gated sodium/potassium channels

Muscarinic Receptors
(G Protein-linked)

- M1
- M2
- M3

Gq-protein-linked:
- ↑ IP3
- ↑ DAG
- ↑ Ca++

Gi-protein-linked:
- ↓ cAMP

q-odd; i-even

Review Lectures 20 & 21 on signal transduction

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Nicotinic cholinergic receptors: Location and function in ANS

Nicotinic receptor subtype	Location	Function/ Effect	Signaling mechanism
Nicotinic neuronal subtype **N_N** receptor	**Autonomic ganglia**	**Postganglionic neuron activation**	Ligand-gated, Na+/K+ channels
	Adrenal medulla	**Chromaffin cells release catecholamines to bloodstream**	

Muscarinic cholinergic receptors: Location and function in ANS

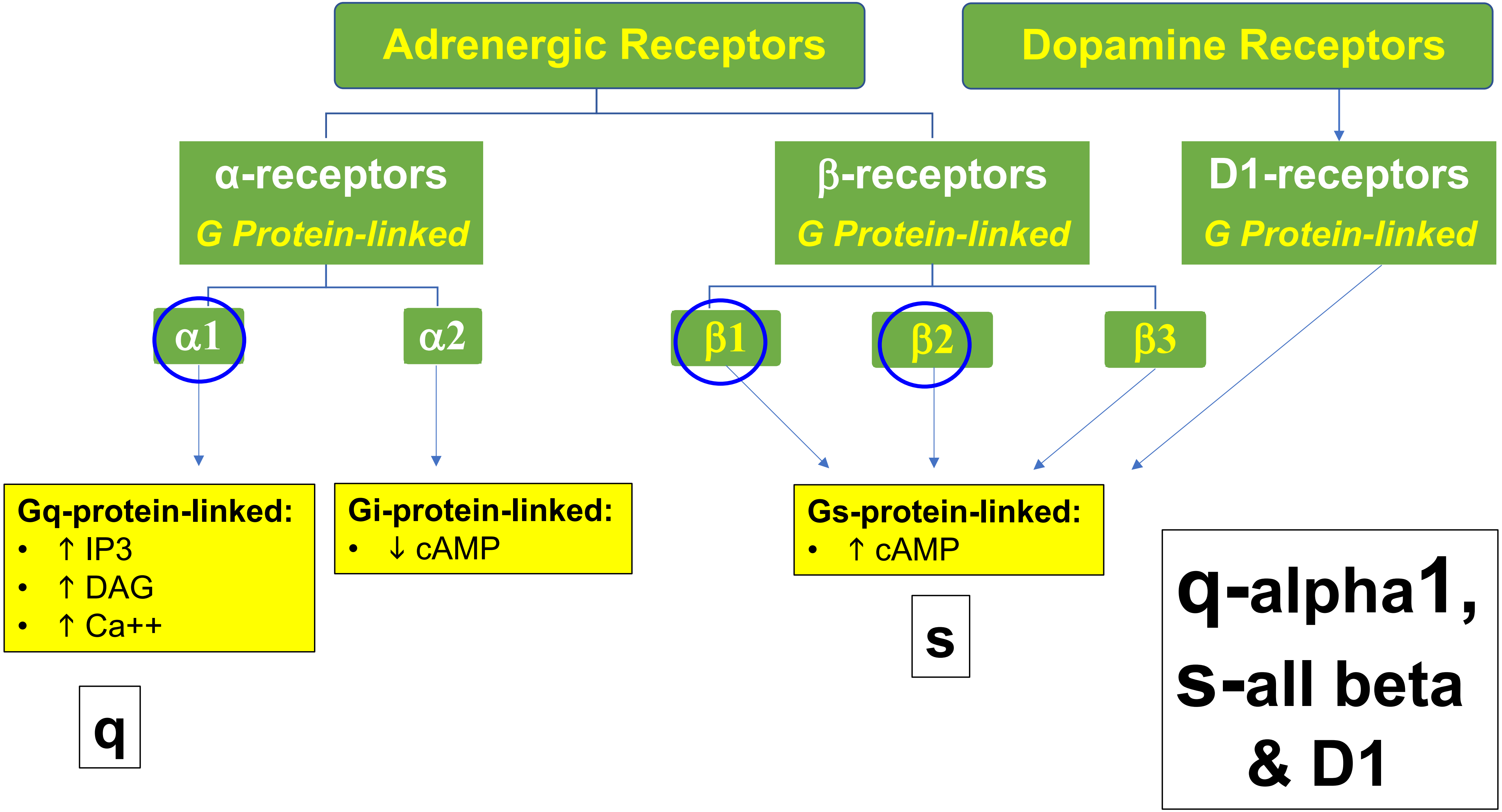
Muscarinic receptor subtype	**Location**	**Function/ Effect**	**Signaling mechanism**
M1 receptor	Gut wall neurons	Stimulates enteric nervous system	Gq-protein-linked: • ↑ IP3 • ↑ DAG • ↑ Ca++
M2 receptor	Presynaptic nerve terminal (neurons)	Inhibition of Ach release	**Gi-protein-linked:** • **↓ cAMP**
	Heart (mainly in the atria)	↓ **Heart rate** ↓ **Contractility of atria**	

Example: Stimulation of the vagus nerve (parasympathetic nerve) to heart slows heart rate, velocity of conduction and atrial contraction through M2

Muscarinic cholinergic receptors: Location and function

Muscarinic receptor subtype	Location	Function/ Effect	Signalling mechanism
M3 receptor	**Exocrine glands**	↑ Exocrine gland secretions (lacrimal, sweat, salivary, gastric acid secretions)	**Gq-protein-linked:** • ↑ IP3 • ↑ DAG • ↑ Ca++
	Smooth muscle:		
	Gastrointestinal smooth muscle	↑ Gut peristalsis	
	Detrusor (urinary bladder) muscle	↑ Bladder contraction	
	EYE: Sphincter pupillae (circular) muscle and ciliary muscle	Pupillary constriction and ↑ curvature of lens (accommodation)	
	Bronchial smooth muscle	Bronchial constriction	
	Endothelial cells	Endothelium NO-mediated vasodilation	

***M3** is the predominant muscarinic cholinergic receptor in many organs*


Adrenergic Receptors
Dopamine Receptors
α-receptors
G Protein-linked
β-receptors
G Protein-linked
D1-receptors
G Protein-linked
α1
α2
β1
β2
β3
Gq-protein-linked:
• ↑ IP3
• ↑ DAG
• ↑ Ca++
Gi-protein-linked:
• ↓ cAMP
Gs-protein-linked:
• ↑ cAMP
q
s
q-alpha1,
s-all beta
& D1

α- Adrenergic receptors: Location and function in ANS

α-receptor subtype	Location	Function/ Effect	Signaling mechanism
Alpha-1 receptors	**Smooth muscle of most blood vessels**	Vasoconstriction	**Gq-protein-linked:** • ↑ IP3 • ↑ DAG • ↑ Ca++
	EYE: dilator pupillae (radial) muscle	Pupillary dilation	
	Arrector pili muscle of skin	Piloerection	

β- Adrenergic receptors: Location and function

β-receptor subtype	Location	Function/ Effect	Signaling mechanism
β1 receptors	**Heart**	↑ Heart rate ↑ Contractility	**Gs-protein-linked:** **↑ cAMP**
	Kidney (Juxtaglomerular cells)	↑ Renin release	
	Ciliary epithelium	↑ Aqueous humor production	
β2 receptors	**Bronchial smooth muscle**	Bronchodilation	
	Blood vessels of skeletal muscles	Vasodilation in skeletal muscles	
	Uterine smooth muscle (myometrium)	↓ Uterine contraction	
	Ciliary epithelium	↑ Aqueous humor production	
β3 receptors	Brown adipose tissue	Thermogenesis and ↑ Lipolysis	

Nitric oxide-containing nerve fibers in Parasympathetic system induce vascular smooth muscle relaxation

Nitric oxide is found in certain **nonadrenergic noncholinergic (NANC)** fibers of the PSNS that innervate the erectile tissues of the penis. Stimulation of these nerves release nitric oxide (NO) and cause penile erection during sex.

Nitric oxide-containing nerve fibers in PSNS induce vascular smooth muscle relaxation

Nitric oxide is found in certain **nonadrenergic noncholinergic (NANC)** fibers of the PSNS that innervate the erectile tissues of the penis. Stimulation of these nerves cause penile erection during sex.

Stimulation of the parasympathetic nerve to erectile tissue in penis(NANC fibers)

↓

Nitric Oxide (NO) release as neurotransmitter

↓

Stimulation of guanylyl cyclase and cGMP production

↓

Smooth muscle relaxation and penile erection

Erectile tissue

ACh binding to muscarinic receptors can also induce similar NO release in this and other tissues

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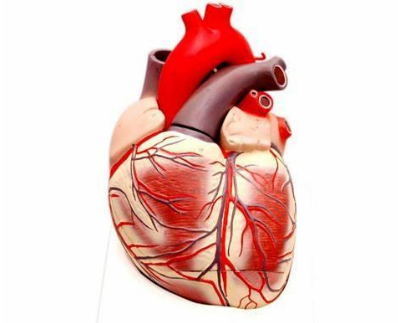
Dual innervation: Rule and Purpose

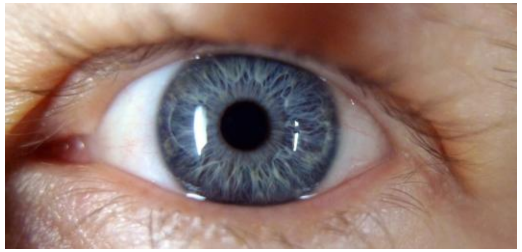
- Many internal organs are innervated by both branches of the ANS **(Eye, heart, lung, gut, and bladder)**.
- SNS and PNS exhibit **ANTAGONISTIC** effects (noncomplementary to each other)
- Dual innervation and antagonism contributes to the cooperative homeostatic regulation by the organ/system.

Sympathetic stimulation
Norepinephrine β1 receptors
↑ HEART RATE



Parasympathetic stimulation
Acetylcholine M2 receptors
↓ HEART RATE

Sympathetic stimulation
Norepinephrine α1 receptors
↑ PUPILLARY SIZE



Parasympathetic stimulation
Acetylcholine M3 receptors
↓ PUPILLARY SIZE

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Tone, Predominant tone of ANS and its importance

- Autonomic tone is the background rate of activity of the ANS in some organs ***at rest***.
- This activity regulates the involuntary functions in the human body that contribute to the homeostasis.
- Predominant tone means that one of the divisions of the ANS has a ***stronger influence*** on the function of that organ at rest.
- Loss of predominant tone results in dysfunction of the organ and system.

Effector organ	Predominant Tone	Effect of LOSS of predominant tone
Blood vessels in skin and gut	Sympathetic	Hypotension (low BP)
SA node in heart	Parasympathetic	↑ Heart Rate (tachycardia)
Glands and muscles in gut	Parasympathetic	Decrease motility & secretion
Iris in the Eye	Parasympathetic	Dilated pupil (mydriasis)

Flipped Classroom Required Reading Material

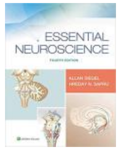
1. Watch Video Recording for Flipped Classroom:
Click below link to watch video:
https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=24a753a9-158c-405d-a0bc-af8c003032f1

2. Read the following related REQUIRED READING material in the recommended Physiology textbook, chapter 6-Autonomic Nervous System,
Under heading, “AUTONOMIC INTEGRATION”. Use link below to reach here:
https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253086161&bookid=3211

Learning objectives linked are:

- SOM.MK.II.BPM1.1.FTM.3.PHYS.0047 Autonomic functions of the eye.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0048 Describe the components and functions of the baroreflex. Explain the clinical implications of baroreflex dysfunction.

Other Recommended readings



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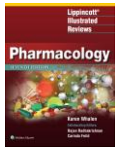

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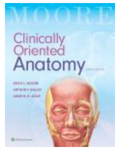

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Any Questions ?

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